IIoT and Cloud Computing (CS5005)

Lecture – 18

Application Protocols for IoT

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Application Protocols for IoT

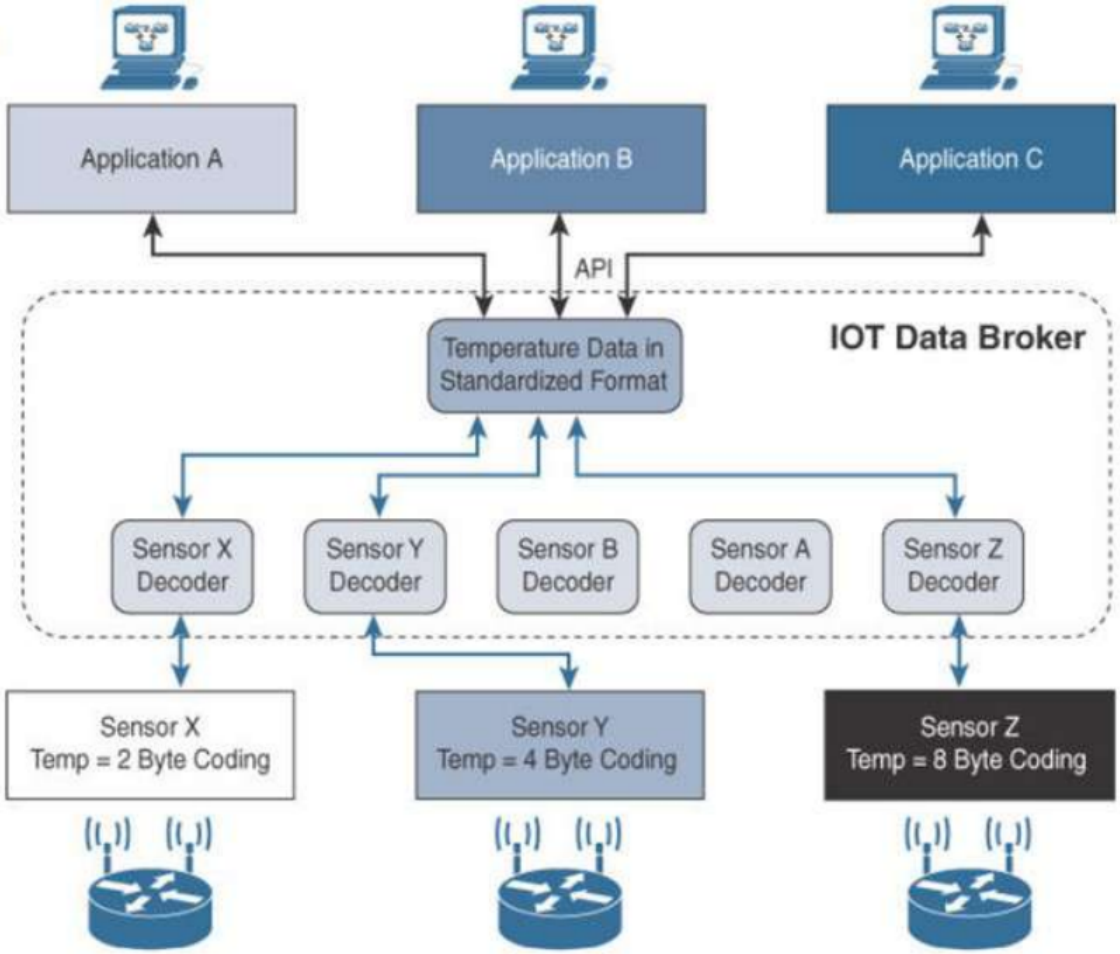
- Transport Layer: IP-based networks use
  - TCP (Connection-Oriented): A session is established between the source and destination before exchanging data.
  - UDP (Connectionless): Data can be quickly sent between source and destination, but with no guarantee of delivery.

Application Protocols for IoT

- IoT Application Transport Methods:
  - Generic Web-based Protocols (conn: HTTP/HTTPS, data: XML/JSON)
  - Message Queuing Telemetry Transport (MQTT)
  - Constrained Application Protocol (CoAP)

Application Protocols for IoT

- IoT Data Broker



Web-Based Protocol: HTTP

- HTTP stands for Hypertext Transfer Protocol. HTTP protocol designed to transfer information between computers over WWW (World Wide Web).
- The data can be (transferred between computers over the Internet)
  - document
  - file
  - image
  - video
- HTTP (Browser) sends a request to a HTTP server.
- The server responds to the message.

HTTP

- The default port is TCP 80.
- Connection Less:
- HTTP is media independent:
- HTTP is stateless: